3RD EXPERIMENT

1ST QUESTION

```
mysql> CREATE DATABASE E3;
Query OK, 1 row affected (0.02 sec)

mysql> USE E3;
Database changed
mysql> CREATE TABLE STUDENTS (StudentID INT PRIMARY KEY, Name VARCHAR(50) NOT NULL, Branch VARCHAR(50));
Query OK, 0 rows affected (0.24 sec)

mysql> CREATE TABLE COURSES (CourseID INT PRIMARY KEY, CourseName VARCHAR(100) NOT NULL);
Query OK, 0 rows affected (0.05 sec)
```

```
mysql> CREATE TABLE TEACHERS (TeacherID INT PRIMARY KEY, Name VARCHAR(50) NOT NULL, Salary DECIMAL(10,2), CourseID INT, FOREIGN KEY (CourseID) REFERENCES CO
URSES(CourseID));
Query OK, 0 rows affected (2.46 sec)

mysql> CREATE TABLE TESTS (TestID INT PRIMARY KEY, TestName VARCHAR(50) NOT NULL, TotalMarks INT);
Query OK, 0 rows affected (0.04 sec)

mysql> CREATE TABLE SCORES (ScoreID INT PRIMARY KEY, Marks INT, StudentID INT, FOREIGN KEY (StudentID) REFERENCES STUDENTS(StudentID), CourseID INT, FOREIGN
 KEY (CourseID) REFERENCES COURSES(CourseID),TestID INT, FOREIGN KEY (TestID) REFERENCES TESTS(TestId));
Query OK, 0 rows affected (2.76 sec)
```

```
mysql> INSERT INTO STUDENTS (StudentID, Name, Branch) VALUES (1, 'RAHUL', 'CSE'), (2, 'PRIYA', 'ECE'), (3, 'AMIT', 'ME'), (4, 'SNEHA', 'CSE'), (5, 'VIKRAM',
 'IT');
Query OK, 5 rows affected (0.05 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> INSERT INTO COURSES (CourseID, CourseName) VALUES (101, 'DATABASE MANAGEMENT SYSTEMS'), (102, 'OPERATING SYSTEMS'), (103, 'COMPUTER NETWORKS'), (104,
 'DATA STRUCTURES'), (105, 'SOFTWARE ENGINEERING');
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> INSERT INTO TEACHERS (TeacherID, Name, Salary, CourseID) VALUES (1, 'Dr. Anjali Mehta',  75000.00, 101),
    -> (2, 'Prof. Rajesh Verma',82000.00, 102),
    -> (3, 'Dr. Kavita Joshi',  78000.00, 103),
    -> (4, 'Prof. Suresh Nair', 85000.00, 104),
    -> (5, 'Dr. Meena Iyer',    79000.00, 105);
Query OK, 5 rows affected (0.02 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> INSERT INTO TESTS (TestID, TestName, TotalMarks) VALUES (1, 'Mid Term',      50),
    -> (2, 'End Semester',  100),
    -> (3, 'Quiz 1',         20),
    -> (4, 'Assignment',     30),
    -> (5, 'Practical Exam', 40);
Query OK, 5 rows affected (0.02 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> INSERT INTO SCORES (ScoreID, StudentID, CourseID, Marks, TestID) VALUES (1, 1, 101, 42, 1),
    -> (2, 1, 101, 88, 2),
    -> (3, 2, 102, 45, 1),
    -> (4, 3, 103, 78, 2),
    -> (5, 4, 104, 19, 3),
    -> (6, 5, 105, 28, 4),
    -> (7, 2, 101, 35, 5),
    -> (8, 3, 104, 92, 2);
Query OK, 8 rows affected (0.01 sec)
Records: 8  Duplicates: 0  Warnings: 0
```

2ND QUESTION

```
mysql> SELECT *FROM STUDENTS;
+-------------+---------+---------+
| StudentID | Name    | Branch |
+-------------+---------+---------+
|           1 | RAHUL   | CSE     |
|           2 | PRIYA   | ECE     |
|           3 | AMIT    | ME      |
|           4 | SNEHA   | CSE     |
|           5 | VIKRAM  | IT      |
+-------------+---------+---------+
5 rows in set (0.01 sec)
```

```
mysql> SELECT *FROM COURSES;
+----------+-------------------------------+
| CourseID | CourseName                    |
+----------+-------------------------------+
|      101 | DATABASE MANAGEMENT SYSTEMS   |
|      102 | OPERATING SYSTEMS             |
|      103 | COMPUTER NETWORKS             |
|      104 | DATA STRUCTURES               |
|      105 | SOFTWARE ENGINEERING          |
+----------+-------------------------------+
5 rows in set (0.00 sec)

mysql> SELECT *FROM TEACHERS;
+-----------+--------------------+----------+----------+
| TeacherID | Name               | Salary   | CourseID |
+-----------+--------------------+----------+----------+
|         1 | Dr. Anjali Mehta   | 75000.00 |      101 |
|         2 | Prof. Rajesh Verma | 82000.00 |      102 |
|         3 | Dr. Kavita Joshi   | 78000.00 |      103 |
|         4 | Prof. Suresh Nair  | 85000.00 |      104 |
|         5 | Dr. Meena Iyer     | 79000.00 |      105 |
+-----------+--------------------+----------+----------+
5 rows in set (0.00 sec)

mysql> SELECT *FROM TESTS;
+--------+----------------+------------+
| TestID | TestName       | TotalMarks |
+--------+----------------+------------+
|      1 | Mid Term       |         50 |
|      2 | End Semester   |        100 |
|      3 | Quiz 1         |         20 |
|      4 | Assignment     |         30 |
|      5 | Practical Exam |         40 |
+--------+----------------+------------+
5 rows in set (0.00 sec)
```

```
mysql> SELECT *FROM SCORES;
+---------+-------+-----------+----------+--------+
| ScoreID | Marks | StudentID | CourseID | TestID |
+---------+-------+-----------+----------+--------+
|       1 |    42 |         1 |      101 |      1 |
|       2 |    88 |         1 |      101 |      2 |
|       3 |    45 |         2 |      102 |      1 |
|       4 |    78 |         3 |      103 |      2 |
|       5 |    19 |         4 |      104 |      3 |
|       6 |    28 |         5 |      105 |      4 |
|       7 |    35 |         2 |      101 |      5 |
|       8 |    92 |         3 |      104 |      2 |
+---------+-------+-----------+----------+--------+
8 rows in set (0.00 sec)
```